

EGS PILLAY ENGINEERING COLLEGE

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

SYNTRONIX'26

PAPER PRESENTATION – SUGGESTED TOPICS

1. Explainable AI: Making AI Decisions Understandable
2. Generative AI and the Future of Software Development
3. AI-Powered Predictive Analytics for Real-World Applications
4. Federated Learning for Privacy-Preserving AI
5. Multimodal AI: Combining Text, Vision and Speech
6. Edge AI: Bringing Intelligence Closer to Devices
7. Reinforcement Learning for Autonomous Decision Making
8. AI Agents and Autonomous Task Automation
9. Bias and Fairness in Machine Learning Models
10. Synthetic Data Generation for AI Training
11. Zero Trust Architecture for Modern Cybersecurity
12. AI-Driven Threat Detection and Cyber Defense
13. Blockchain-Based Identity and Access Management
14. Post-Quantum Cryptography: Preparing for the Quantum Era
15. Privacy-Preserving Machine Learning
16. Deepfake Detection Using Artificial Intelligence
17. Ransomware Detection and Prevention Techniques
18. Passwordless Authentication and Modern Identity Security
19. Security Challenges in Internet of Things
20. Adversarial Attacks Against Machine Learning Systems
21. Quantum Computing: Opportunities and Challenges for CSE
22. Internet of Things and Intelligent Connected Systems
23. Digital Twins: Virtual Models for Real-World Systems
24. Spatial Computing and the Future of Human-Computer Interaction
25. Extended Reality: Computing Beyond the Physical World
26. 5G/6G Networks and Intelligent Computing
27. Brain-Computer Interfaces and Computational Intelligence
28. Swarm Intelligence for Distributed Problem Solving
29. Neuromorphic Computing: Computing Inspired by the Brain

30. Autonomous Systems and Intelligent Robotics
31. Serverless Computing: The Next Evolution of Cloud Applications
32. Cloud-Native Application Architecture
33. Edge Computing vs Cloud Computing
34. Sustainable Computing and Green Data Centers
35. Digital Infrastructure for Scalable AI Systems
36. Microservices Architecture for Large-Scale Applications
37. DevSecOps: Integrating Security into Software Development
38. Self-Healing Software Systems Using AI
39. Low-Code and No-Code Development: Future of Software Engineering
40. AI-Assisted Software Testing and Debugging
41. Big Data Analytics for Intelligent Decision Making
42. Knowledge Graphs and Intelligent Information Retrieval
43. Graph Neural Networks for Complex Data Analysis
44. Real-Time Data Processing with Stream Computing
45. Data Mining for Personalized Recommendation Systems
46. Computer Vision for Smart Applications
47. Natural Language Processing for Multilingual Computing
48. Sentiment Analysis Using Large Language Models
49. Human-Centered AI for Inclusive Technology
50. Ethical AI: Building Responsible and Trustworthy Computing Systems